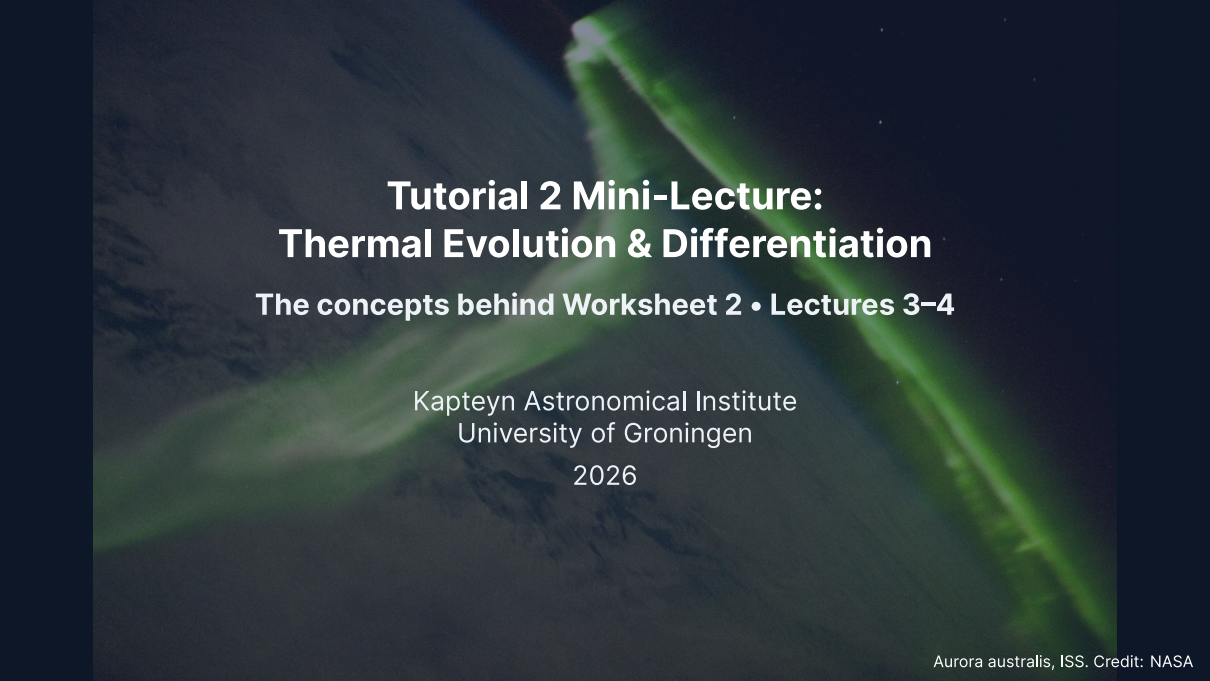


A photograph of the aurora australis (southern lights) as seen from the International Space Station (ISS). The image shows vibrant green and blue light curtains against the black background of space. The lights appear as long, flowing ribbons of color, with some brighter, more intense green sections and others in a softer blue. The perspective is from above, looking down at the glowing atmosphere of Earth.

Tutorial 2 Mini-Lecture: Thermal Evolution & Differentiation

The concepts behind Worksheet 2 • Lectures 3–4

Kapteyn Astronomical Institute
University of Groningen

2026

What Worksheet 2 trains

Five problems, all built on Lectures 3–4. The physics you need, per problem:

1. **The radiogenic clock:** isotope table to heating power, now and at formation.
2. **Conduction fails, convection takes over:** $\tau = L^2/\kappa$, Ra, Nu.
3. **A minimal thermal evolution model:** heat in minus heat out, on paper and in Python.
4. **Building a core and dating it:** Stokes settling, partitioning, Hf-W ages.
5. **Dynamos and shields:** magnetic Reynolds number, standoff distance.

Every problem has the shape and level of one exam question. Problem 3(c) is the only part that needs a computer.

The radiogenic clock

Specific heating = concentration $\times$ heat per kg of isotope, summed:

$$h = \sum_i c_i h_i, \quad c_i(\text{past}) = c_i 2^{t/t_{1/2}}$$

- Today: $\approx 4.9 \times 10^{-12} \text{ W kg}^{-1}$ of BSE rock $\Rightarrow \approx 20 \text{ TW}$.
- At formation (4.5 Gyr ago): 4.7 times more, $\approx 92 \text{ TW}$.

Two clocks, two eras

^{26}Al ($t_{1/2} = 0.72 \text{ Myr}$) out-heats everything for the first few Myr, then dies: planetesimals melt on the short clock, planets stay warm on the long one.

The isotope table is the whole calculation: concentrations, half-lives, heat rates. Everything else is bookkeeping.

Conduction fails, convection takes over

Conductive cooling time and convective vigour:

$$\tau = \frac{L^2}{\kappa},$$

$$Ra = \frac{\rho g \alpha \Delta T D^3}{\kappa \mu}$$

- Moon: $\tau \approx 10^{11}$ yr $\gg$ age.
- Only bodies ≤ 400 km cool conductively in 4.57 Gyr.
- Mantle: $Ra \approx 5 \times 10^7 \gg Ra_c \approx 1708$.

How much faster is convection?

$Nu = (Ra/Ra_c)^{1/3} \approx 32$: thirtyfold. And since $Nu \propto \Delta T^{1/3}$, the flux scales as $\Delta T^{4/3}$: hotter mantles shed heat disproportionately fast.

L^2 decides who conducts; Ra decides who convects; Nu says how vigorously.

A minimal thermal evolution model

One line of energy conservation:

$$C \frac{dT}{dt} = H(t) - Q(T)$$

- $H(t)$: radiogenic heating, a decaying exponential.
- $Q(T) \propto T^{4/3}$: convective loss (from $Nu \propto Ra^{1/3}$).
- You complete two lines of Python and press run.

What the curve does

Start hot but under-radiating: the interior *warms* until Q catches H (~1 Gyr), then tracks the dying heat source down. Today: the model cools at ~170 K per Gyr (the real mantle nearer 50–100), Urey ratio $H/Q \approx 0.4$.

Planets self-regulate: the $T^{4/3}$ thermostat ties the heat loss to the heat supply.

Building a core and dating it

Three tools, one story:

$$v_{\text{Stokes}} = \frac{2}{9} \frac{\Delta \rho g r^2}{\mu}, \quad \frac{C^{\text{sil}}}{C^{\text{bulk}}} = \frac{1}{1 + x(D - 1)}$$

- Iron rain crosses a magma ocean in **weeks**.
- Siderophile elements ($D \gg 1$) vanish into the core; their leftover excess records the **late veneer**.

Core formation is fast, chemically selective, and datable: three properties, three worksheet parts.

The Hf-W stopwatch

$^{182}\text{Hf} \rightarrow ^{182}\text{W}$, $t_{1/2} = 8.9$ Myr: Hf stays in the mantle, W leaves with the metal. The mantle's $\epsilon^{182}\text{W}$ excess dates core formation, until the parent goes extinct at ~45 Myr.

Dynamos and shields

Dynamo feasibility and its shield:

$$\boxed{R_m = \frac{UL}{\eta} \gtrsim 10-100}, \quad r_{\text{mp}} = R_p \left(\frac{B_0^2}{\mu_0 \rho_{\text{sw}} v_{\text{sw}}^2} \right)^{1/6}$$

- Even Ganymede's 300 km shell clears the bar.
- $\tau_{\text{ohm}} = L^2/\eta \sim \text{kyr}$: an observed field is regenerated *now*.

The sixth root

A tenfold solar-wind surge moves Earth's magnetopause only from 10 to $6.8 R_{\oplus}$. Weak dependence is the shield's strength.

R_m asks whether a dynamo can run; the pressure balance says how far its shield reaches.

Working the worksheet

- **Show your algebra.** The exam asks for the steps, not only the number.
- Carry **4 significant figures** through intermediate steps; round at the end.
- Use the **checkpoints** ("show that ...") to verify your work and to keep moving if a step resists.
- Qualitative parts want **2 to 3 sentences** of physics, not an essay.
- For Problem 3(c), download `thermal_evolution.py` from the worksheets page; any Python 3 with numpy and matplotlib works.

Worksheet and full solutions: `ips.formingworlds.space/worksheets.html`.
We discuss questions, not presentations of the solutions.